

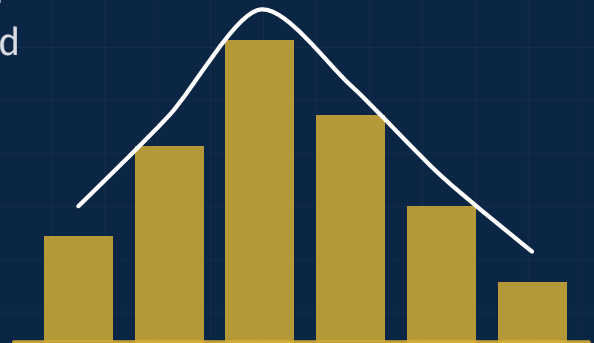


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CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 3

Cumulative Frequency

Add up as you go, plot at the top of each class, draw a smooth curve — then read across for the median and quartiles, and up for “how many”.



3

VARIANTS

55

PAST-PAPER PARTS

24

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Table & graph

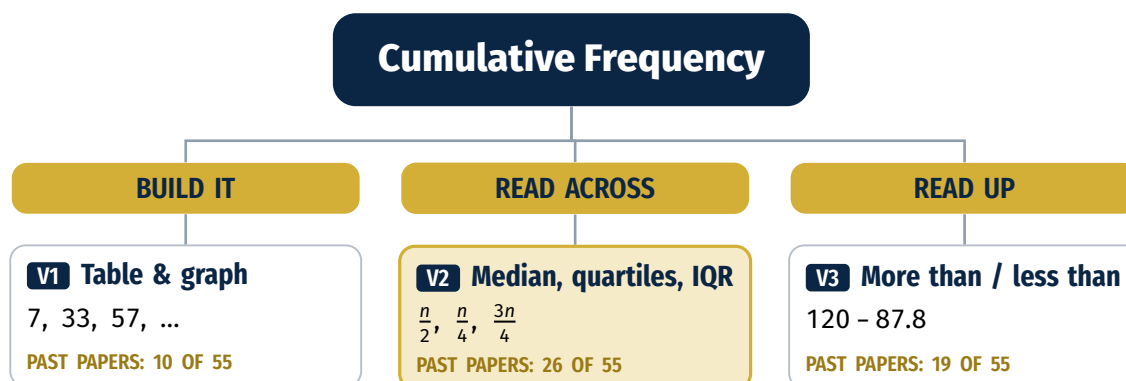
V2 Median, quartiles, IQR

V3 More than / less than

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Cumulative frequency at a glance



Your notes


THE KEY HABIT

PLOT AT THE UPPER BOUND

(20, 33): by 20 minutes, 33 people had finished.

★ How often does each variant come up?

20 past-paper questions (Jan 2020 → Nov 2025), about 55 parts; 24 practice parts — grouped by type.

	PAST PAPERS (OUT OF 55)	PRACTICE PARTS
V1 Table, graph	10	15
V2 Median, quartiles, IQR	26	3
V3 More than, less than	19	6

WHAT THIS TELLS YOU On almost every paper, one long question (Q11–14, 5–8 marks): complete the table, plot, read the median/IQR, then “how many more than ...”. Every part is a routine skill.

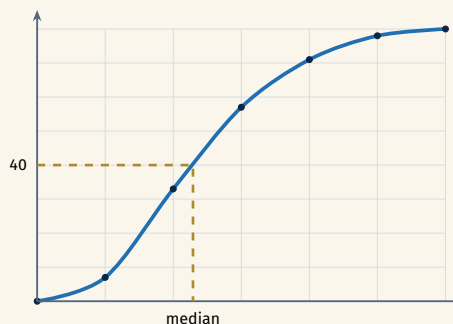
BEFORE YOU START

The big idea

**A running total.**

Cumulative frequency = how many values are **up to** each point. It only goes **up**, and ends at the total.

Read **across** from a count to find a value (median, quartiles); read **up** from a value to find a count.


Where to read across (n values)
Median

$$\frac{n}{2}$$

80 people: read at 40.

Lower quartile

$$\frac{n}{4}$$

At 20.

Upper quartile

$$\frac{3n}{4}$$

At 60. IQR = $Q_3 - Q_1$.

Reading up
Less than x

read up at x

The cf there.

More than x

$n - \text{cf at } x$

Subtract from the total.

**CALCULATOR FIRST, THEN CHECK**

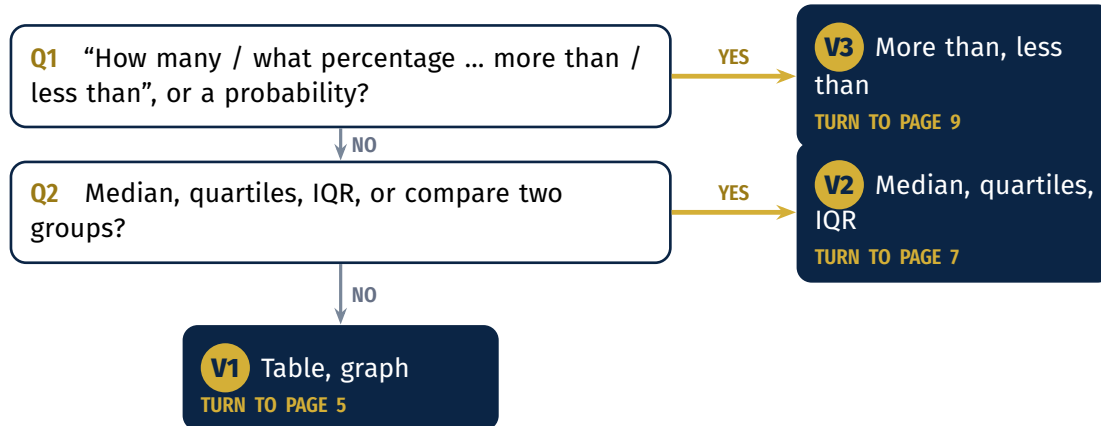
Draw the reading lines on the graph in pencil — they earn the method mark. Answers are estimates: a small range is accepted.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1

Cumulative Frequency Table & Graph

Running totals; plot each at the class's upper bound; join with a smooth curve.

SEEN IN PAST PAPERS

10 / 55

PRACTICE QUESTIONS: 15

 Your notes

Came up in: Jan 2021 P1HR Q11 · Jan 2022 P1H Q12 · Jun 2022 P2HR Q11 · Jun 2022 P2H Q13 · Jun 2024 P2HR Q11 · Jun 2024 P2H Q13

? What is it?

In plain words

Keep a **running total** of the frequencies. The last one equals the total.

Plot each running total at the **upper bound** of its class, start at the lowest bound with 0, and join with a **smooth curve** (or straight lines).

IT LOOKS LIKE

 $0 < t \leq 10 : 7$
 $10 < t \leq 20 : 26$

class	f	cf	point
$0 < t \leq 10$	7	7	(10, 7)
$10 < t \leq 20$	26	33	(20, 33)
$20 < t \leq 30$	24	57	(30, 57)
$30 < t \leq 40$	14	71	(40, 71)
$40 < t \leq 50$	7	78	(50, 78)
$50 < t \leq 60$	2	80	(60, 80)

How do I do it?

1 Running total

Add each frequency to the last cf.

2 Check

Last cf = total.

3 Plot

(upper bound, cf); start at (lowest bound, 0).

4 Join

Smooth curve through all points.

PLOT AT THE UPPER BOUND Not the midpoint.

V1 Cumulative Frequency Table & Graph

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Complete the cumulative frequency table” → **Running totals**

“draw a cumulative frequency graph for your table” → **(upper bound, cf), smooth curve**

A table already cumulative
($10 < a \leq 30$: 36) → **Plot straight away**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 1H • Q12(a)(b) • 4 marks

80 customers: $0 < t \leq 10$: 7, $10 < t \leq 20$: 26, $20 < t \leq 30$: 24, $30 < t \leq 40$: 14, $40 < t \leq 50$: 7, $50 < t \leq 60$: 2. Complete the cumulative frequency table and draw the graph.

STEP 1 $7, 7 + 26 = 33, 33 + 24 = 57$ ◀ running total

STEP 2 $57 + 14 = 71, 71 + 7 = 78, 78 + 2 = 80$ ◀ running total

STEP 3 $(0, 0), (10, 7), (20, 33), \dots, (60, 80)$ ◀ upper bounds

STEP 4 Join with a smooth S-shaped curve ◀ draw

ANSWER cf: 7, 33, 57, 71, 78, 80

✓ ends at 80 ✓

WATCH OUT

✗ plotting at the midpoints (5, 15, ...)

✓ at the upper bounds (10, 20, ...)

By 10 minutes, 7 had finished.

EXAM TIP

The last cumulative frequency must equal the total — a free check.

Check yourself

calculator first • answers on page 11

1.1

Jun 2024 P2H

Distances: 7, 17, 18, 14, 10, 4 in classes of 10 km from 0 to 60. Cumulative frequencies?

Answer:

1.2

Jun 2022 P2H

Ages: 0–20: 7, then 25, 20, 14, 8, 6 in classes of 10 years. Cumulative frequencies?

Answer:

1.3

Dr Eslam

Frequencies 6, 14, 20, 7, 3 for $0 < t \leq 10, \dots, 40 < t \leq 50$. Cumulative frequencies?

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Median, Quartiles & IQR

Read across at $\frac{n}{2}$, $\frac{n}{4}$, $\frac{3n}{4}$; $IQR = Q_3 - Q_1$.

SEEN IN PAST PAPERS

26 / 55

PRACTICE QUESTIONS: 3



Your notes

Came up in: Jan 2020 P1HR Q12 • Jan 2020 P1H Q12 • Jan 2021 P1H Q11 • May 2021 P2H Q12 • Jan 2022 P2HR Q14 • Jun 2022 P2HR Q11 • May 2023 P1HR Q14 • May 2023 P1H Q12 • Nov 2023 P1H Q13 • Jun 2024 P2HR Q11 • Nov 2024 P2H Q13 • Jun 2025 P2H Q13 • Nov 2025 P2H Q14

? What is it?

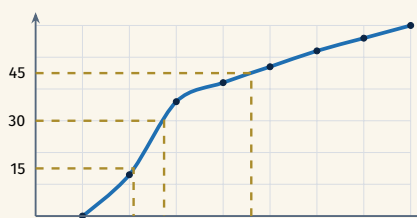
In plain words

Go to the $cf = \frac{n}{2}$ on the vertical axis, **across** to the curve, **down** to the value: the **median**.

Same at $\frac{n}{4}$ (Q_1) and $\frac{3n}{4}$ (Q_3). Compare groups: **median** for “on average”, **IQR** for “consistent / spread out”.

IT LOOKS LIKE

60 people: read at 30 15 45



How do I do it?

1 Positions

 $\frac{n}{2}, \frac{n}{4}, \frac{3n}{4}$

2 Across

From each cf to the curve.

3 Down

Read the value on the horizontal axis.

4 IQR / compare

$Q_3 - Q_1$; compare median and IQR.

$n/4, n/2, 3n/4$ The most common variant.

V2 Median, Quartiles & IQR

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Use the graph to find an estimate for the median" → **Across at $\frac{n}{2}$**

"estimate for the interquartile range" → **Across at $\frac{n}{4}$ and $\frac{3n}{4}$**

"Compare the waiting times at A and B" → **Median (average) and IQR (consistency)**

WORKED EXAMPLE **PAST PAPER**

June 2022 • Paper 2HR • Q11(b)(c) • 3 marks

Ages of 60 people (cumulative): up to 20: 13, 30: 36, 40: 42, 50: 47, 60: 52, 70: 56, 80: 60. Estimate the median and the IQR.

STEP 1 median at 30 : $20 + \frac{30-13}{36-13} \times 10 = 27.4$ ◀ between 20 and 30

STEP 2 Q_1 at 15 : $20 + \frac{2}{23} \times 10 = 20.9$ ◀ lower quartile

STEP 3 Q_3 at 45 : $40 + \frac{3}{5} \times 10 = 46$ ◀ upper quartile

STEP 4 $46 - 20.9 = 25.1$ ◀ IQR

ANSWER median ≈ 27.4 years, IQR ≈ 25.1 years ✓ reading lines on the graph agree ✓

WATCH OUT

✗ median at cf 30.5 or the 30th value of a list

✓ read at $\frac{n}{2} = 30$

For a graph, use $\frac{n}{2}$ exactly.

EXAM TIP

"Compare": one sentence on the average (median), one on the spread (IQR), both in context.

Check yourself

calculator first • answers on page 11

2.1

Jan 2022 P1H

80 customers: cf 7, 33, 57, 71, 78, 80 at 10, 20, ..., 60 minutes. Estimate the median.

Answer:

2.2

Jun 2022 P2H

80 people: cf 7, 32, 52, 66, 74, 80 at 20, 30, ..., 70 years. Estimate the median.

Answer:

2.3

Jun 2024 P2HR

60 workers: cf 6, 17, 27, 42, 53, 60 at 5, 10, ..., 30 hours. Estimate the IQR.

Answer:

V3**VARIANT 3****More Than / Less Than**

Read up at the value; “more than” is total – reading; then percentage or probability.

SEEN IN PAST PAPERS

19 / 55

PRACTICE QUESTIONS: 6


 Your notes

Came up in: Jan 2020 P1H Q12(b) • Jan 2021 P1HR Q11(d) • Jan 2021 P1H Q11(b) • Nov 2021 P1H Q11 • Jan 2022 P1H Q12(d) • Jan 2022 P2HR Q14 • Jun 2022 P2HR Q11(d) • Jun 2022 P2H Q13(d) • Jun 2024 P2HR Q11 • Jun 2024 P2H Q13(d) • Jun 2025 P1HR Q12

? What is it?**In plain words**

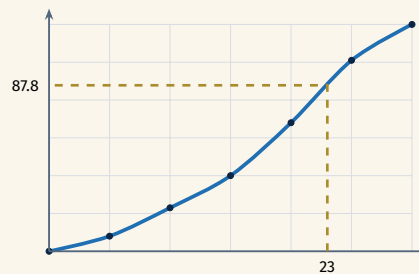
Go **up** from the value to the curve, then across: that many are **less than** it.

More than: total – reading. Then a fraction, percentage or probability of the total. Backwards (“25 spent more than W”): read across at total – 25.

IT LOOKS LIKE

longer than 23 min

more than W

**How do I do it?****1 Up**

From the value to the curve.

2 Across

Read the cf: “less than”.

3 More than

Total – cf.

4 Convert $\frac{\text{count}}{\text{total}}, \times 100$ for %.**MORE THAN = TOTAL – READING** Show the reading line.

V3 More Than / Less Than

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"percentage ... who waited longer than 23 minutes" → 120 - cf at 23, then $\div 120$

"25 workers spent more than W hours" → Across at $60 - 25 = 35$

"probability that $5 < H \leq 10$ " → cf at 10 - cf at 5, $\div$ total

**WORKED EXAMPLE** **PAST PAPER**

January 2021 • Paper 1HR • Q11(d) • 2 marks

120 people; cf 8, 23, 40, 68, 101, 120 at 5, 10, 15, 20, 25, 30 minutes.
Estimate the percentage who waited longer than 23 minutes.

STEP 1 cf at 23 $\approx 68 + \frac{3}{5} \times 33 = 87.8$ ◀ read up at 23

STEP 2 $120 - 87.8 = 32.2$ ◀ more than 23

STEP 3 $\frac{32.2}{120} \times 100$ ◀ percentage

STEP 4 ≈ 26.8 ◀

ANSWER about 27%

✓ between 20 (68 done) and 25 (101 done) ✓

**WATCH OUT**

× giving 87.8 (less than 23)

✓ $120 - 87.8$

"Longer than" is the top part.

**EXAM TIP**

Read to the nearest small square;
answers within a range get the marks.

**Check yourself**

calculator first • answers on page 11

3.1

Jun 2024 P2HR

60 workers: cf 6, 17, 27, 42, 53, 60 at 5, 10, ..., 30 hours. 25 workers spent more than W hours. Find W.

Answer:

3.2

Jun 2024 P2HR

Same workers.
Probability that $5 < H \leq 10$.

Answer:

3.3

Jun 2024 P2H

70 teachers: cf 7, 24, 42, 56, 66, 70 at 10, ..., 60 km. How many travel more than 46 km?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked in Python (check_answers.py); readings use straight lines between plotted points.

1.1

7, 24, 42, 56, 66, 70

1.2

7, 32, 52, 66, 74, 80

1.3

6, 20, 40, 47, 50

2.1

about 23 min

2.2

about 34 years

2.3

about 12.3 hours

3.1 $W \approx 17.7$ **3.2** $\frac{11}{60}$ **3.3**

about 8

THE WHOLE TOPIC ON ONE PAGE

Cumulative frequency cheat sheet



Your notes
V1 Table, graph 10/55

7, 33, 57, ...

Running total; plot at upper bounds.

V2 Median, IQR 26/55 $\frac{n}{4}, \frac{n}{2}, \frac{3n}{4}$

Read across, then down.

V3 More / less 19/55

120 – 87.8

Read up; total – reading.

**TOP 3 LOST MARKS**

1. plotting at midpoints ×
2. “more than” given as the reading ×
3. no reading lines drawn ×

BOTTOM LINE Running total • plot at the upper bound • across at $\frac{n}{4}, \frac{n}{2}, \frac{3n}{4}$ •
more than = total – reading.